faculty of science
department of mathematics

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MATH 100, Spring 2026
Final Exam
April 20, 2026, 12:00 – 15:00

SFU Email:	MRA175 @SFU.CA	Signature:	MR

First name (please write as legibly as possible within the boxes)

M	e	l	o	d	i	e										

Last name

R	o	u	s	s	e	l										

Student number (9 digits)

3	0	1	6	6	2	7	9	6								

Please Read the Following Instructions

1. Do not open this booklet until told to do so.
2. **Using blockletters and dark pen,** write your name, SFU student number, and email ID in the space provided.
3. This exam is **180 minutes** long and consists of **20 questions**.
4. You may use the last page for extra space. **You must indicate below the question if any of your work is written elsewhere.**
5. To receive full credit for written questions, your solution must be complete and well presented. You must cross out any work that you do not wish to be graded.
6. **Calculators are not allowed.** You should leave answers as "calculator ready" expressions (such as $\sqrt{3}$ or π); however, answers should be simplified as much as possible.
7. No books, notes, smartphones, or any device capable of connecting to the internet shall be within the reach of a student during the examination.
8. **During the examination, copying from, communicating with, or deliberately exposing written papers to the view of other examinees is forbidden.**

Question:	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Total
Points:	3	4	3	5	4	4	5	4	3	4	5	4	3	4	3	5	4	4	4	5	80

Potentially Useful Identities and Formulas

Addition and Subtraction Trigonometric Identities:

$$\cos(\alpha+\beta) = \cos(\alpha)\cos(\beta) - \sin(\alpha)\sin(\beta)$$
$$\cos(\alpha-\beta) = \cos(\alpha)\cos(\beta) + \sin(\alpha)\sin(\beta)$$
$$\sin(\alpha+\beta) = \sin(\alpha)\cos(\beta) + \cos(\alpha)\sin(\beta)$$
$$\sin(\alpha-\beta) = \sin(\alpha)\cos(\beta) - \cos(\alpha)\sin(\beta)$$

Double Angle Trigonometric Identities:

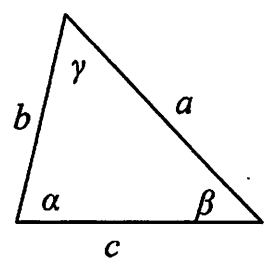
$$\sin(2\theta) = 2\sin(\theta)\cos(\theta)$$
$$\cos(2\theta) = \cos^2(\theta) - \sin^2(\theta) = 1 - 2\sin^2(\theta) = 2\cos^2(\theta) - 1$$

Half Angle Trigonometric Identities:

$$\sin\left(\frac{\alpha}{2}\right) = \pm\sqrt{\frac{1-\cos(\alpha)}{2}}$$
$$\cos\left(\frac{\alpha}{2}\right) = \pm\sqrt{\frac{1+\cos(\alpha)}{2}}$$

Laws of Sines and Cosines:

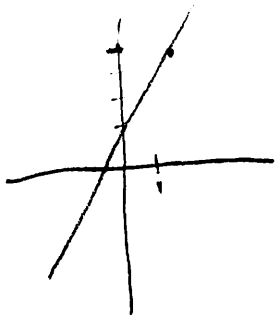
Consider a triangle with angles and opposite sides as shown:



$$\frac{\sin(\alpha)}{a} = \frac{\sin(\beta)}{b} = \frac{\sin(\gamma)}{c}$$

$$a^2 = b^2 + c^2 - 2bc\cos(\alpha)$$
$$b^2 = a^2 + c^2 - 2ac\cos(\beta)$$
$$c^2 = a^2 + b^2 - 2ab\cos(\gamma)$$

[3] 1. A line passes through $(\overset{x}{1}, \overset{y}{4})$ and is perpendicular to $3x - 2y = 6$. Find the equation of this line.



$3(x) - 2(4) = 6$

$3(x) - 8 = 6$

$+8 \quad +8$

$\frac{3(x)}{3} = \frac{14}{3}$

$x = \frac{14}{3}$

$3(1) - 2(y) = 6$

$3 - 2(y) = 6$

$-3 \quad -3$

$\frac{-2(y)}{2} = \frac{3}{2}$

$y = \frac{3}{2}$

$f(x) = \frac{14}{3}(x) + \frac{3}{2}(y)$

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[4] 2. Let $f(x) = \sqrt{x-3}$ and $g(x) = x^2 + 1$. Find the simplified expression for $(g \circ f)(x)$ and state its domain in interval notation.

$(g \circ f)(x)$

$g(f(x))$

$g(\sqrt{x-3})$

$g(\sqrt{(x^2+1)-3})$

$g(x+1)-3)$

$x \neq 0$

$x \neq -1$

$[-\infty, 0] \cup [0, -1] \cup [0, \infty]$

[3] 3. Starting with the parent function $f(x) = |x|$, write the equation for $g(x)$ that results from a horizontal shift right by 2 units, a vertical stretch by a factor of 3, and a reflection over the x-axis.

$f(x) = |x|$

$g(x) = (-3x+2)$

2 units R
stretch x 3
reflect over x

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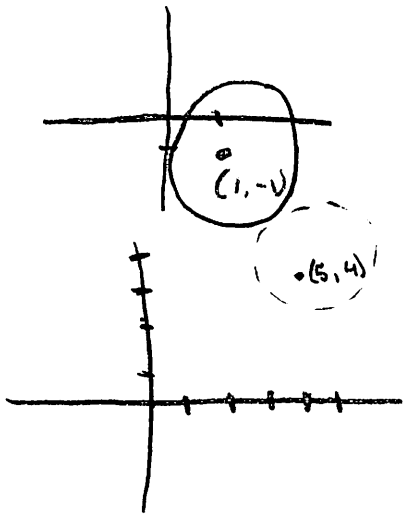
[5] 4. Find the inverse $f^{-1}(x)$ for $f(x) = \dfrac{2x-5}{3x+1}$.

$f^{-1}(x) = \dfrac{3y+1}{2y-5}$

[4] 5. Write the equation of the circle in standard form with centre $(1, -1)$ passing through the point $(5, 4)$.

$x^2 + y^2 = 1$ = unit circle

$f(x) = (x+5) + 5$



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[4] 6. Factor $P(x) = x^3 - 4x^2 - 7x + 10$ completely, given that $(x - 1)$ is one of its factors. State all real zeros.

$$
\begin{array}{r}
x^2 + 5x + 8 \\
(x-1)\overline{)\,x^3 - 4x^2 - 7x + 10} \\
-x^3 - 1 \qquad\quad \downarrow \qquad \downarrow \\
\hline
5x^2 - 7x + 10 \\
-5x^2 - 1 \qquad \downarrow \\
\hline
8x + 10 \\
-8x - 1 \\
\hline
9
\end{array}
$$

$x^2 + 5x + 8$

$(x+3)(x+2)$

$\boxed{\text{The zeros are } -3 \text{ and } -2}$

[5] 7. For the function $R(x) = \dfrac{2x^2 - 8}{x^2 - x - 6}$, identify any holes, vertical asymptotes, and horizontal asymptotes.

Vertical Asym	Horizontal Asym	Hole?
$y = \dfrac{2(0)^2 - 8}{(0)^2 - (0) - 6}$ $y = \dfrac{-8}{-6}$ $y = \dfrac{-8}{-6}$ is a vertical asymptote	?	?

[4] 8. A population of bacteria doubles every 4 hours. If the initial population is 500, write an exponential model for $P(t)$, where t is measured in hours. Leave your answer in calculator-ready format.

x 2 every 4 hrs.

$P_0 = 500$ $T = \text{hrs}$

$P(t) = (P_0)^{rT}$

$P(t) = (500)^{(2)(T)}$

$P(t) = (500)^{2t}$

[3] 9. Use properties of logarithms to expand $\log\left(\frac{x^3\sqrt{y}}{z^2}\right)$. Express in terms of $\log(x)$, $\log(y)$, and $\log(z)$.

$\log\left((x^3 + \sqrt{y}) - z^2\right)$

$\log(x)$, $\log(y)$, $\log(z)$

$\log_3(x)$, $\log_2(y)$, $\log_2(z)$

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[4] 10. Solve for x:

$$\ln(x) + \ln(x-3) = \ln(4)$$

$\ln(x) + \ln(x-3) = \ln(4)$

$\ln(x^2 - 3) = \ln(4)$

$+3 \qquad +3$

$\sqrt{\ln(x^2)} = \sqrt{\ln(4)} + 3$

$x = 4 + 3$

$x = 7$

[5] 11. A car bought for \$25,000 depreciates at a rate of 12% per year. After how many years will the car be worth exactly \$10,000? Express your answer in calculator-ready format.

$r = 12\% \ (0.12)$

$P_0 = 25000$

$T = 12$ months $= 1$ yr

P_0 = price initial

P_n = price new

$P(t) = 25000^{t(0.12)}$

$10000 = 25000^{t(0.12)}$

$\log 10000 = t(0.12) \log(25000)$

$\frac{\log(10000)}{\log(25000)} = \frac{t(0.12)}{(0.12)}$

$\frac{\left(\frac{\log(10000)}{\log(25000)}\right)}{(0.12)} = t$

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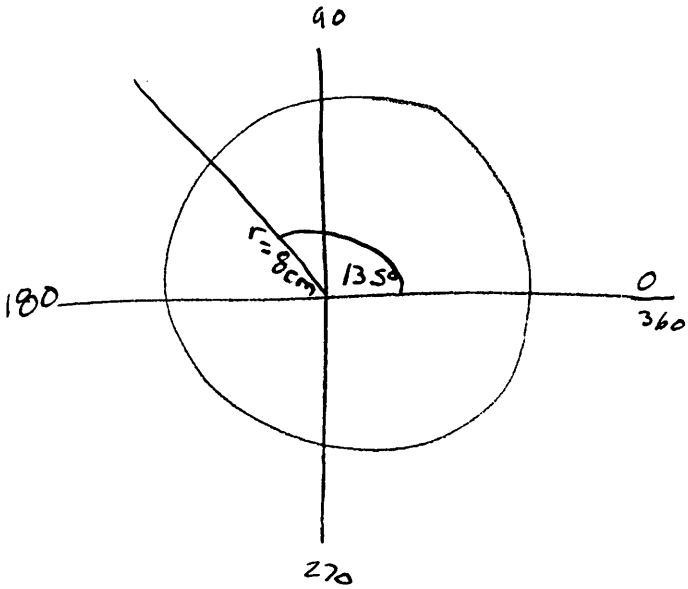
[4] 12. Find the exact length of an arc intercepted by a central angle of 135° in a circle with a radius of 8 cm.

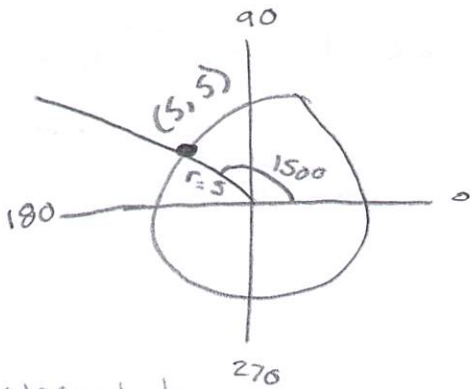


arc length $= \frac{135}{8}$

[3] 13. A point P is on a circle of radius 5 at an angle of $\theta = \frac{5\pi}{6}$. Determine the exact (x, y) coordinates of point P.

$r = 5 \qquad \theta = \frac{5\pi}{6}$

$\frac{5\pi}{6} \times \frac{180}{\pi} = \frac{900}{6} = 150°$



unit circle $= x^2 + y^2 = 1$
(radius = 1)

$P = (r = 5)$

New circle enlarged by 5 times

So $x = 5$ and $y = 5 = (5, 5)$

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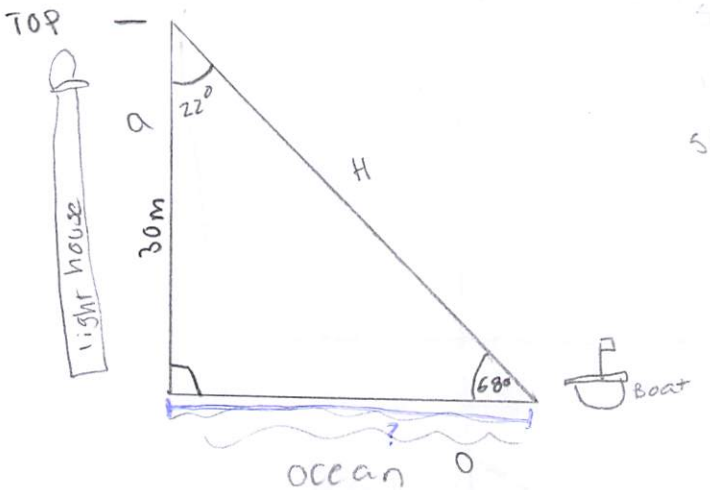
[4] 14. From the top of a 30 m lighthouse, the angle of depression to a boat in the ocean is 22°. How far is the boat from the base of the lighthouse? Leave your answer in calculator-ready format.



$\sin(22°)$

[3] 15. Simplify the expression $(\sec\theta - \tan\theta)(\sec\theta + \tan\theta)$ using fundamental identities.

$(\sec\theta - \tan\theta)(\sec\theta + \tan\theta)$

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[5] 16. State the amplitude, period, phase shift, and midline for $f(x) = -4\sin(2x - \pi) + 3$.

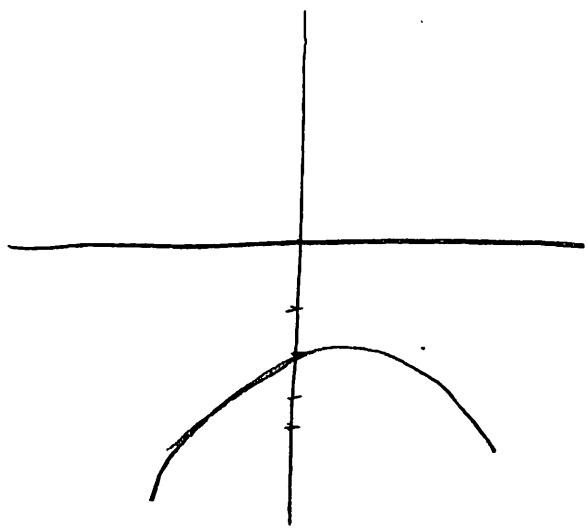
reflect 4 down comp π right 3 up

Amplitude : -4

period :

Phase shift : 3 up

midline : -2



[4] 17. Find the exact value of $\sin\left(\cos^{-1}\left(-\frac{1}{2}\right)\right)$.

$\sin\left(\cos\left(\frac{2}{1}\right)\right)$

$\sin(\cos(2))$

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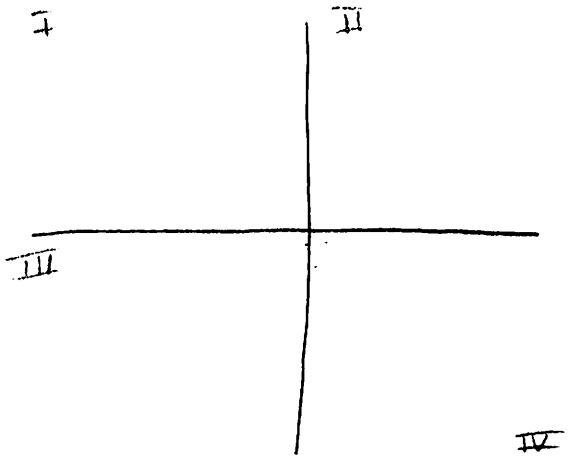
[4] 18. Use an addition or subtraction identity to find the exact value of $\cos(75°)$.

$\cos(a+B) = \cos(a) + \cos(B) - \sin(a) + \sin(B)$

[4] 19. If $\sin\theta = \frac{3}{5}$ and θ is in Quadrant II, find the exact value of $\sin(2\theta)$.



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[5] 20. In triangle ABC, side $a = 10$, side $b = 12$, and the contained angle $C = 60°$. Find the exact length of side c.

$c^2 = a^2 + b^2 - 2ab\cos(\gamma)$

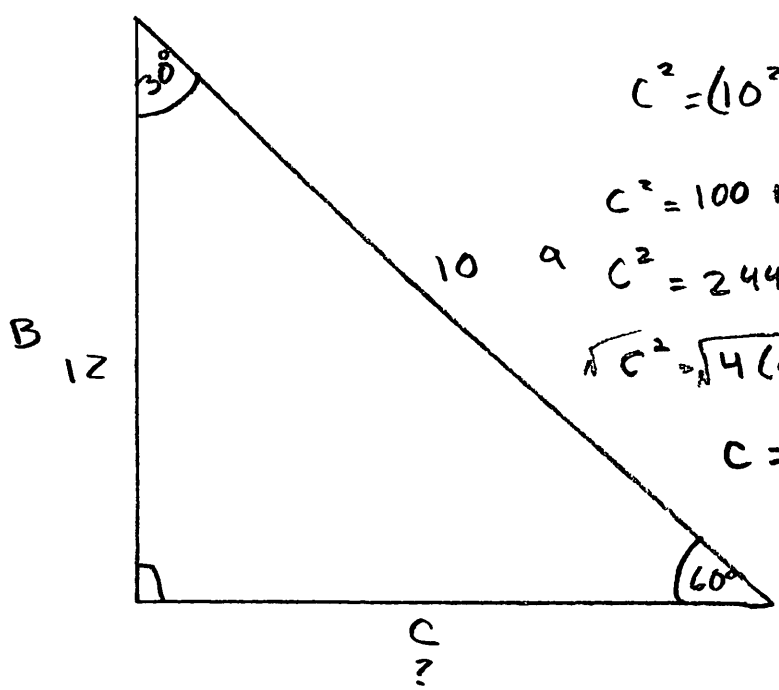
$c^2 = (10^2 + 12^2) - 2(12 \times 10)\cos(30°)$

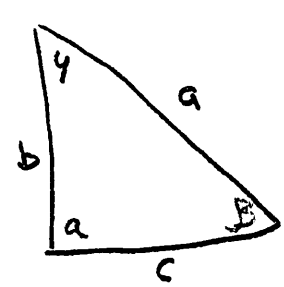
$c^2 = 100 + 144 - (240)\cos(30°)$

$c^2 = 244 - 240(\cos(30°))$

$\sqrt{c^2} = \sqrt{4(\cos(30°))}$

$c = \sqrt{4(\cos(30°))}$





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